

PROGRAMS NO: 15

```
text = input("Enter ciphertext: ")
freq = "etaoinshrdlucmfwyvbgkjqxz"

results = []

for k in range(26):
    p = ''.join(chr((ord(c)-97-k)%26+97) if c.isalpha() else c for c in text.lower())
    score = sum(p.count(x) for x in freq)
    results.append((score, p))

results.sort(reverse=True)

n = int(input("How many plaintexts? "))

for i, (score, p) in enumerate(results[:n], 1):
    print(i, ":", p)
```

Output:

```
Enter ciphertext: koor zruog
How many plaintexts? 10
1 : zwddg ogjdv
2 : yvccf nficu
3 : xubbe mehbt
4 : wtaad ldgas
5 : vszzc kcfzr
6 : uryyb jbeyq
7 : tqxxa iadxp
8 : spwwz hzcwo
9 : rovvv gybvn
10 : qnuux fxaum
```